A Parsimonious Two-Factor Asset Pricing Model: Theory and Evidence

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Abstract

This paper proposes a parsimonious two-factor model that outperforms well-known multifactor models in the asset pricing literature. Dubbed the Mean Variance Pricing (MVP) model, it is based on the ZCAPM of Kolari, Liu, and Huang (2021) derived from Black's (1972) famous zero-beta CAPM. The pricing factors are the first and second moments of all stocks in the market on any given day (i.e., cross-sectional mean market returns and cross-sectional total market return dispersion of all stocks). We show that, using different test asset portfolios in out-of-sample cross-sectional regression tests, the goodness-of-fit and statistical significance of factor loadings associated with total market return dispersion outperforms a number of prominent multifactor models. In these tests, the MVP model performs similar to the more complex EM ZCAPM previously used to estimate the ZCAPM. Also, graphs comparing out-of-sample fitted to realized portfolio returns reveal markedly smaller average mispricing errors for the MVP model relative to other models.

Keywords: asset pricing, multifactor models, zero-beta CAPM, ZCAPM
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1 Introduction

Kolari, Liu, and Huang (2021) developed the ZCAPM asset pricing model, which was derived as a special case of Black's (1972) famous zero-beta CAPM. The general equilibrium ZCAPM is comprised of two factors: (1) cross-sectional mean market returns, and (2) cross-sectional standard deviation of returns of stocks in the market (i.e., total market return dispersion). Both factors are readily available for the U.S. stock market (for example). To empirically estimate the ZCAPM, the authors specify a relatively complex regression model that employs the expectation-maximization (EM) algorithm. While EM regression methods are often used in the hard sciences, most economics and finance researchers are unfamiliar with them. Extensive empirical tests showed that their EM ZCAPM regression approach consistently outperformed prominent multifactor models by Fama and French (1992, 1993, 2015, 2018) and others. Based on two-step Fama and MacBeth (1973) tests, out-of-sample cross-sectional regression analyses showed that the market price of loadings associated with total market return dispersion (viz., so-called *zeta risk*) were highly significant with t-statistics ranging from 3 to 6 in tests of different portfolios and individual stocks. Factor loadings in other multifactor models were less significant in all tests. Estimated R^2 values well exceeded other models also.[1]

Despite impressive cross-sectional regression results on an out-of-sample basis, researchers have been reticent to adopt the EM ZCAPM in their empirical asset pricing work. The main problem is difficulty in understanding the iterative EM algorithm, which has led to hesitancy in accepting the exceptional results of the EM ZCAPM.[2] In this paper, in an effort to overcome EM ZCAPM issues, we propose a simpler Mean Variance Pricing (MVP) model to estimate the ZCAPM via standard least squares regression methods. Common statistical software programs, including Microsoft Excel, can be used to estimate the MVP model. In forthcoming analyses, using different test asset portfolios for U.S. stocks, we show that the MVP model consistently outperforms prominent multifactor models and is comparable to the EM ZCAPM. We conclude that the MVP model is a simple yet powerful general equilibrium model that warrants adoption by researchers in the field of asset pricing.

[1] For further evidence, see Kolari, Huang, Butt, and Liao (2022), Kolari, Huang, Liu, and Liao (2022, 2025, 2026), Kolari, Liu, and Pynnonen (2024), Liu, Kolari, and Huang (2012), and Liu (2013).

[2] The authors make available Matlab, R, and Python programs to estimate the EM ZCAPM at GitHub (https://Github.com/zcapm).

2 ZCAPM Review

2.1 Theoretical ZCAPM

The general equilibrium Capital Asset Pricing Model (CAPM) of Sharpe (1964), Lintner (1965), Mossin (1966), and Treynor (1961, 1962) was a landmark in asset pricing. The CAPM was derived from the Markowitz (1952, 1959) mean-variance parabola. Expected asset returns were specified as a function of the riskless rate plus the market factor and associated beta risk. Unfortunately, early empirical tests did not support the CAPM.

Subsequently, Black (1972) allowed borrowing at a risky rate greater than the riskless rate, or $E(R_Z) > R_f$, and short selling to derive the zero-beta CAPM:

$$E(R_i) \quad = \quad E(R_Z) + \beta_i[E(R_M) - E(R_Z)], \tag{1}$$

where $E(R_M)$ = the expected market portfolio return; $E(R_Z)$ is the expected return on the uncorrelated (orthogonal) zero-beta portfolio; and β_i = beta risk. Roll (1980) observed that Black's model can be generalized to any combination of an efficient portfolio I and its zero-beta portfolio counterpart ZI:[3]

$$E(R_i) \quad = \quad E(R_{ZI}) + \beta_{iI}[E(R_I) - E(R_{ZI})]. \tag{2}$$

If investors can own riskless assets (but not borrow at the riskless rate), Black (1972, p. 454) obtained:

$$E(R_i) - R_f = \beta_{iI}[E(R_I) - R_f] + \beta_{iZI}[E(R_{ZI}) - R_f], \tag{3}$$

where β_{iI} and β_{iZI} are factor loadings related to expected excess returns of portfolios I and ZI, respectively.

Kolari, Liu and Huang (2021) derived a special case of Black's zero-beta CAPM. Figure 1 illustrates the location of two portfolios with equal total risk denoted I^* and ZI^* on the parabola with approximate expected returns:

$$E(R_{I^*}) \quad \approx \quad E(R_m) + \sigma_m \tag{4}$$

$$E(R_{ZI^*}) \quad \approx \quad E(R_m) - \sigma_m, \tag{5}$$

[3]See also Copeland and Weston (1980, p. 184).

Substituting these expected returns into Black's relation (2), they derived the ZCAPM as follows:

$$\begin{aligned} E(R_i) &= E(R_{ZI^*}) + \beta_{iI^*}[E(R_{I^*}) - E(R_{ZI^*})] \\ &= E(R_m) - \sigma_m + \beta_{iI^*}\{[E(R_m) + \sigma_m] - [E(R_m) - \sigma_m]\} \\ &= E(R_m) + (2\beta_{iI^*} - 1)\sigma_m \\ E(R_i) &= E(R_m) + Z^*_{im}\sigma_m, \end{aligned} \tag{6}$$

where $Z^*_{im} = 2\beta_{iI^*} - 1$. Assuming a riskless asset, they obtained the theoretical ZCAPM:

$$E(R_i) - R_f = \beta_{im}[E(R_m) - R_f] + Z^*_{im}\sigma_m, \tag{7}$$

where β_{im} is beta risk; and Z^*_{im} = zeta risk related to the cross-sectional total market return dispersion of all assets in the market. On any given day t, these two factors can be readily computed for all stocks in the market on a cross-sectional basis as the first and second moments of returns. Note that the expected returns of portfolios I^* and ZI^* require that zeta risk be positive and negative, respectively. Importantly, the authors mathematically proved that the vertical width or span of the mean-variance parabola of Markowitz is closely related to the cross-sectional standard deviation of returns of all assets in the market, or σ_m. As the parabola's vertical width changes over time, assets located in the upper (lower) half of the parabola would experience increasing (decreasing) expected returns as σ_m increases, and vice versa when σ_m decreases. Additionally, the cross-sectional mean market return denoted $E(R_m)$ (e.g., typically proxied by the value-weighted CRSP index return) is located near the axis of symmetry in the middle of the parabola, rather than near the efficient frontier as believed by many researchers.

To empirically estimate the theoretical ZCAPM, the authors specified the following regression model:

$$R_{it} - R_{ft} = \alpha_i + \beta_{im}(R_{mt} - R_{ft}) + Z_{im}D_{it}\sigma_{mt} + u_{it}, \quad t = 1, \ldots, T, \tag{8}$$

where a dummy signal variable $D_{it} = +1$ or -1 captures potential positive and negative zeta risk, $u_{it} \sim$ iid $\mathrm{N}(0, \sigma_i^2)$, and other notation is as before. Since D_{it} is unknown, they employ an expectation-maximization (EM) algorithm to estimate the probability p_i that $D_{it} = +1$.[4] Their EM ZCAPM is as follows:

$$R_{it} - R_{ft} = \beta_{im}(R_{mt} - R_{ft}) + Z^*_{im}\sigma_{mt} + u_{it}, \quad t = 1, \ldots, T, \tag{9}$$

[4]See Dempster Laird and Rubin (1977), Jones and McLachlan (1990), McLachlan and Peel (2000), McLachlan and Krishnan (2008), and others. The EM algorithm is widely used in the hard sciences to estimate hidden variables in regression analyses.

where $Z^*_{im} = Z_{im}(2p_i - 1)$. The authors do not include an intercept (or α_i) as it tended to increase residual error variance. For empirical tests using U.S. stocks, the value-weighted CRSP index was used to proxy the average market return R_{mt} on day t, the Treasury bill rate proxied R_{ft}, and the value-weighted cross-sectional standard deviation of stock returns in the CRSP index was computed for σ_{mt}. Many researchers consider the latter market return dispersion factor to be representative of various macroeconomic shocks, such as business cycles, economic uncertainty, market volatility, etc.[5]

Some studies have added total market return dispersion to the CAPM to test a two-factor model, including Jiang (2010), Demirer and Jategaonkar (2013), and Garcia Mantilla-Garcia and Martellini (2014). However, their results were mixed and much less significant than EM ZCAPM results. Cross-sectional regression tests showed that the EM ZCAPM consistently outperformed leading multifactor models by Fama and French (1992, 1993, 2015, 2018) and others. Goodness-of-fit as measured by R^2 values were markedly higher than other models. Also, the estimated market price of zeta risk loadings was economically meaningful with t-statistics greater than 3, which well exceeded those associated with other often-used factor loadings (e.g., size, value, momentum, etc.). Additionally, average mispricing errors were much smaller than those of other models. Extensive tests using different test asset portfolios and individual stocks confirmed these results. Also, previously cited research by the authors corroborated the dominance of the EM ZCAPM compared to other multifactor models.

3 The MVP Model

Our Mean Variance Pricing (MVP) model simplifies the estimation of an empirical ZCAPM. Unlike the complex iterative EM ZCAPM, standard least squares regression methods can be utilized. As such, the MVP model can be estimated with readily available regression software (including Excel).

To derive the MVP model, we consider the possibility that market return dispersion is zero. Under this boundary condition, referring to Figure 1, the parabola collapses to a single line at its axis of symmetry. Now the expected return of all assets equals the expected return of the global minimum variance portfolio, or $E(R_{it}) = E(R_{Gt})\ \forall i$. As proven in the Appendix A, under these conditions, the empirical ZCAPM does not contain an intercept parameter (or alpha). Consequently, we specify the MVP model as follows:

$$R_{it} - R_{ft} = \beta_{im}(R_{mt} - R_{ft}) + Z_{im}\sigma_{mt} + u_{it}, \quad t = 1, \ldots, T, \tag{10}$$

where zeta risk coefficient Z_{im} does not require the estimation of probability p_i by the iterative EM algorithm.

[5]See Loungani, Rush, and Tave (1990), Bekaert and Harvey (1997), Bansal and Yaron (2004), Pastor and Veronesi (2009), and others.

Because there is no intercept (α_i), the in-sample Gibbons, Ross, and Shanken (GRS) (1989) test cannot be used to investigate its validity. For this reason, we test the MVP model via standard out-of-sample Fama-MacBeth cross-sectional regression methods. Previous researchers[6] have argued that out-of-sample analyses provide more reliable tests of model validity than in-sample tests. Moreover, they are akin to investable strategies wherein buy or sell decisions are made and thereafter returns per unit risk are assessed in the next period.

4 Methodology

4.1 Data

For test asset portfolios in sample period January 1972 to December 2024, we downloaded daily returns from Kenneth French's website[7] for 50 firm characteristic portfolios (viz., 25 size-book-to-market portfolios and 25 operating profitability-capital investment portfolios). Also, 48 industry portfolios are downloaded and combined with these 50 portfolios to form 98 diversified portfolios. Lastly, in the sample period July 1972 to December 2021, we gathered 286 anomaly portfolios comprised of 153 plus 133 portfolios from the online websites of Jensen, Kelly, and Pederson (2023) and Chen and Zimmerman (2022), respectively.

The market factor is computed as the daily excess return of the value-weighted CRSP index over the Treasury bill rate. The market return dispersion factor is the value-weighted cross-sectional standard deviation of stocks' returns on day t. To compare the MVP model to other models, we download from French's website daily return series for the following factors: size (SMB), value (HML), momentum (MOM), profitability (RMW), and capital investment (CMA).

Descriptive statistics for factors in the sample period are shown in Table 1. The market return dispersion factor (σ_{mt}) is 1.86 percent per day, which is much larger than the market factor ($R_{mt} - R_{ft}$) return at 0.03 percent. In unreported results, their correlation coefficient is only 0.05. By contrast, a number of commonly-used factors exhibit nontrivial correlation coefficients: $\rho(SMB, RMW) = -0.29$, $\rho(HML, MOM) = -0.29$, and $\rho(HML, CMA) = 0.55$. Also, the market factor has some degree of correlation with multifactors HML, MOM, RMW, and CMA (i.e., -0.15, -0.14, -0.22, and -0.34, respectively).

[6] See Simin (2008), Ferson, Nallareddy, and Xie (2013), Barillas and Shanken (2018), Fama and French (2018), Martin and Nagel (2021), and others.

[7] See https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.

4.2 Cross-sectional regression tests

Based on the Fama-MacBeth procedure, we estimate the factor loadings for each model using one year of daily returns and then run an out-of-sample cross-sectional regression of one-month-ahead returns on estimated model factor loadings. This two-step process is rolled forward one month at a time in the sample period. Monthly OLS cross-sectional regression estimates of the market prices of risk, or λ_ks, are averaged and associated t-statistics computed.[8] To examine average mispricing errors for each model, we generate monthly realized and predicted (one-month-ahead) returns for test asset portfolios in all sample months. Following Jagannathan and Wang (1996) and Lettau and Ludvigson (2001), cross-sectional OLS regression of average realized returns on average predicted returns is estimated to obtain an R^2 goodness-of-fit value.[9]

5 Empirical Results

In Table 2, we report out-of-sample (one-month-ahead) cross-sectional regression results for the following models: CAPM, MVP-A (with intercept), MVP (no intercept), and EM ZCAPM. For the 50 firm characteristic portfolios, Panel A shows that the MVP model and EM ZCAPM have higher R^2 values at 79 percent and 91 percent compared to the MVP-A model at 41 percent and CAPM at 37 percent, respectively. However, when industry portfolios are added to these portfolios, as shown in Panel B for 98 diversified portfolios, the latter models have little or no explanatory power with R^2 at most 4 percent. Even so, the MVP model maintains its explanatory power with R^2 at 79 percent, and the EM ZCAPM performs well at 64 percent. Turning to the 286 anomaly portfolios in Panel C, again only ZCAPM-based models exhibit strong goodness-of-fit at 84 percent and 81 percent for the MVP model and EM ZCAPM, respectively, and the CAPM and MVP-A model have no explanatory power.

In terms of the significance of the estimated market prices of risk, the MVP model and EM ZCAPM have highly significant zeta risk loadings (i.e., $\hat{\lambda}_{zeta}$) with t-statistics in the range of 3.50 to 7.28 across the three different test asset portfolios.[10] For the anomaly portfolios, the t-statistics exceed 7, which is very high compared to extant asset pricing literature. Note that market factor loadings are normally insignificant across the different models (except the EM ZCAPM for anomaly portfolios). Also, zeta risk loadings are generally insignificant in the MVP-A model (except in the diversified portfolios with an unexpected negative sign).

[8] See Cochrane (1996), Ferson (2019), Giglio and Xiu (2021), and others. For the MVP model and EM ZCAPM, the zeta risk coefficient or factor loading is multiplied by 21 to rescale the loading from a daily to monthly basis as in Kolari, Liu, and Huang (2021, p. 127).

[9] In the MVP time-series regression model, the estimated R^2 value is not directly comparable to an OLS estimate of R^2 (i.e., it does not measure the proportion of variance explained). However, out-of-sample cross-sectional regressions are estimated via OLS, such that R^2 values can be clearly interpreted.

[10] A t-statistic greater than 3 is recommended by Harvey, Liu, and Zhu (2016) and Chordia, Goyal, and Saretto (2020) to avoid the risk of false discoveries of factors.

In sum, our results demonstrate that the MVP model performs well across a wide spectrum of test asset portfolios including difficult to explain anomaly portfolios. Despite its simplicity, it performs comparable to the more complex EM ZCAPM.

In Table 3, we document the cross-sectional regression results for the following well-known multifactor models: Fama and French three-, five-, and six-factor models (denoted FF3, FF5, and FF6, respectively) and Carhart four-factor model (denoted C4). These models perform fairly well in terms of R^2 values using the 50 firm characteristic portfolios (Panel A) (i.e., 65 percent R^2 estimate for FF6) but not for diversified portfolios (Panel B) and especially anomaly portfolios (Panel C) (i.e., 8 percent R^2 estimate for FF6). A number of estimated market prices of risk for different factor loadings are significant in Panels A and B (i.e., momentum, profitability, and capital investment factors with t-statistics from 2.0 to 2.9), but only the capital investment factor is significant in Panel C. In general, these renowned models are dominated by the ZCAPM-based models. The former models exhibit better performance using test asset portfolios constructed from the same firm characteristics employed in constructing factors; however, their performance diminishes when exogenous asset portfolios are tested.

5.1 Mispricing error graphs

To study mispricing errors in asset pricing models, Cochrane (1996), Lettau and Ludvigson (2001), and others graphically compare out-of-sample fitted (predicted) to realized (actual) returns. Using 98 diversified portfolios, Figure 2 shows that the FF6 model has somewhat lower average pricing errors than the C4 model (where perfect pricing is along the 45-degree line). For these portfolios, Figure 3 reveals that the EM ZCAPM and MVP model have lower average mispricing errors than the FF6 and C4 models. By casual inspection, the MVP model appears to have the lowest mispricing errors among all models.

Anomaly portfolios are notorious for the difficulty in pricing these long/short portfolios. In line with previous literature, Figure 4 shows that the C4 and FF6 models are almost useless in pricing these portfolios. In sharp contrast, Figure 5 illustrates that the EM ZCAPM and MVP model do an excellent job of pricing anomaly portfolios with relatively small average mispricing errors. Thus, we infer that ZCAPM-based models can price these otherwise anomalous portfolios. Moreover, the least squares MVP model can price these assets as well as the more complex EM ZCAPM.

6 Conclusion

This paper proposed a parsimonious two-factor MVP model based on the ZCAPM of Kolari, Liu, and Huang (2021). These authors specified a complex expectation-maximization (EM) algorithm to estimate an empirical

EM ZCAPM regression model with two macrofactors: (1) the cross-sectional mean market return (related to beta risk), and (2) the cross-sectional standard deviation of returns of all assets in the market (associated with zeta risk). We hypothesized that, if market return dispersion is zero, the mean-variance parabola of Markowitz (1952, 1959) collapses to a straight line. Under these conditions, no intercept is implied in the empirical ZCAPM. Consequently, we specify the MVP model with no intercept, which can be estimated with standard least squares regression methods.

Out-of-sample, cross-sectional regression tests were conducted for different asset pricing models, including prominent multifactor models. Using a variety of test asset portfolios based on U.S. stock returns, we found that ZCAPM-based models consistently outperformed other multifactor models in terms of goodness-of-fit as measured by R^2 values. Strikingly, zeta risk factor loadings in the EM ZCAPM and MVP model were very significant with t-statistics considerably higher than other well-known factors under study. Further tests of average mispricing errors confirmed the outperformance of the ZCAPM-based models compared to other multifactor models. Despite its simplicity, the MVP model yielded the lowest mispricing errors compared to other models. Surprisingly, while other models could not price large numbers of anomaly portfolios, the MVP model did an excellent job of pricing difficult to explain anomaly portfolios.

Future research on the MVP model is recommended in the following areas: (1) applications to other asset classes; (2) tests in different countries; (3) identification of anomaly portfolios; and (4) investment analyses to construct high performing stock portfolios.

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Appendix

A Zero Market Return Dispersion Boundary Condition

This Appendix provides theoretical and empirical discussion of the ZCAPM under the extreme boundary condition wherein the cross-sectional standard deviation of market expected returns (i.e.., total market return dispersion) approaches zero. Dropping the time subscript for convenience, assuming that $\sigma_m = 0$, the feasible set in the mean-variance parabola collapses to a horizontal line representing a constant expected return across various asset volatilities, or $E(R_i) = \mu$ for all assets i. Given that risk-averse investors optimize their utility by minimizing variance, because all assets have identical expected returns, investors will choose the global minimum variance portfolio (G). Also, market beta using portfolio G equals one for all assets.

To solve for G, we minimize the portfolio variance subject to the full-investment constraint:

$$\min_{w} \frac{1}{2} w^T \Sigma w \quad \text{s.t.} \quad w^T \mathbf{1} = 1, \tag{A.1}$$

where w is the vector of portfolio weights, and Σ is the variance-covariance matrix of asset returns. The well-known Lagrangian is:

$$\mathcal{L} = \frac{1}{2} w^T \Sigma w - \lambda (w^T \mathbf{1} - 1). \tag{A.2}$$

Taking the first derivative with respect to weight vector w and setting it to zero yields the first-order condition (FOC):

$$\Sigma w - \lambda \mathbf{1} = \mathbf{0} \implies \Sigma w = \lambda \mathbf{1}, \tag{A.3}$$

where Σw represents the vector of covariances between each individual asset and portfolio G. Hence, the covariance of every asset with G is uniform and equal to the Lagrange multiplier λ:

$$Cov(R_i, R_G) = \lambda \quad \forall i. \tag{A.4}$$

To determine the variance of portfolio G, we pre-multiply the FOC by w^T:

$$w^T \Sigma w = w^T (\lambda \mathbf{1}) = \lambda (w^T \mathbf{1}) \tag{A.5}$$

Since $w^T \mathbf{1} = 1$, we obtain:

$$Var(R_G) = \lambda. \tag{A.6}$$

Given that the optimal market portfolio (denoted m) is G at this boundary state, we substitute portfolio m into the standard beta definition:

$$\beta_{im} = \frac{Cov(R_i, R_m)}{Var(R_m)} = \frac{\lambda}{\lambda} = 1 \quad \forall i. \tag{A.7}$$

This theoretical result maps directly to the following straightforward empirical specification of the ZCAPM:

$$R_i - R_f = \alpha_i + \beta_{im}(R_m - R_f) + Z_{im}\sigma_m + \epsilon_i. \tag{A.8}$$

Under the boundary condition $\sigma_m = 0$, the cross-sectional risk factor drops out ($Z_i\sigma_m = 0$). As proven above, equilibrium forces $\beta_{im} = 1$. Furthermore, because $E(R_i) = E(R_m)$ $\forall i$, the intercept α_i must equal zero. Thus, the empirical ZCAPM reduces to the following zero-intercept form:

$$R_i - R_f = (R_m - R_f) + \epsilon_i. \tag{A.9}$$

While the optimal choice of the efficient frontier shrinks to a single point, individual assets retain unique expected returns due to idiosyncratic error term ϵ_i. Because investors optimally diversify away this firm-specific variance by holding optimal market portfolio m, it is not priced. Thus, as the cross-sectional dispersion σ_m approaches zero, the cross-sectional dispersion of β_{im} is forced to approach zero as well.

Table 1: Descriptive statistics: 1972-2024.

Statistic	$R_m - R_f$	σ_m	*SMB*	*HML*	*MOM*	*RMW*	*CMA*
Mean	0.03	1.86	0.00	0.01	0.03	0.01	0.01
Standard deviation	1.07	0.65	0.58	0.61	0.81	0.42	0.39
Maximum	11.36	10.77	6.08	6.73	7.14	4.57	2.48
Minimum	-17.44	0.68	-11.15	-5.03	-14.37	-2.97	-5.31

Table 2: Out-of-sample cross-sectional regression tests for CAPM-based models and test asset portfolios.

Panel A: 50 firm characteristics				
Model	$\hat{\alpha}$	$\hat{\lambda}_{beta}$	$\hat{\lambda}_{zeta}$	R^2
CAPM	0.97	-0.29		0.37
	(4.20***)	(-1.22)		
MVP-A	0.77	-0.11	-0.02	0.41
	(3.50***)	(-0.46)	(-1.27)	
MVP	0.71	-0.05	0.33	0.79
	(3.26***)	(-0.20)	(3.50***)	
EM ZCAPM	0.68	0.00	0.43	0.91
	(3.07***)	(-0.01)	(4.75***)	
Panel B: 98 diversified portfolios				
Model	$\hat{\alpha}$	$\hat{\lambda}_{beta}$	$\hat{\lambda}_{zeta}$	R^2
CAPM	0.61	0.07		0.01
	(3.12***)	(0.33)		
MVP-A	0.50	0.15	-0.02	0.04
	(2.58***)	(0.67)	(-1.81*)	
MVP	0.34	0.33	0.34	0.70
	(1.82*)	(1.48)	(4.33***)	
EM ZCAPM	0.43	0.25	0.38	0.64
	(2.29**)	(1.16)	(5.08***)	
Panel C: 286 anomaly portfolios				
Model	$\hat{\alpha}$	$\hat{\lambda}_{beta}$	$\hat{\lambda}_{zeta}$	R^2
CAPM	0.31	0.15		0.00
	(16.63***)	(0.62)		
MVP-A	0.30	0.16	0.01	0.02
	(16.81***)	(0.68)	(0.47)	
MVP	0.21	0.32	0.72	0.84
	(10.42***)	(1.44)	(7.28***)	
EM ZCAPM	0.21	0.39	0.69	0.81
	(10.42***)	(1.78*)	(7.06***)	

Note: Asterisks on t-statistics in parentheses indicate the level of statistical significance: *–10 percent, **–5 percent, and ***–1 percent.

Table 3: Out-of-sample cross-sectional regression tests for prominent multifactor models and test asset portfolios.

Panel A: 50 firm characteristics								
Model	$\hat{\alpha}$	$\hat{\lambda}_{beta}$	$\hat{\lambda}_{size}$	$\hat{\lambda}_{value}$	$\hat{\lambda}_{mom}$	$\hat{\lambda}_{profit}$	$\hat{\lambda}_{investment}$	R^2
FF3	0.68	-0.04	0.08	0.20				0.39
	(3.66***)	(-0.21)	(0.70)	(1.68*)				
C4	0.77	-0.14	0.09	0.19	0.26			0.43
	(4.13***)	(-0.64)	(0.73)	(1.62)	(1.33)			
FF5	0.59	0.06	0.13	0.20		0.25	0.20	0.64
	(3.11***)	(0.29)	(1.03)	(1.71*)		(2.87***)	(2.60***)	
FF6	0.68	-0.04	0.13	0.21	0.08	0.25	0.20	0.65
	(3.61***)	(-0.22)	(1.03)	(1.73*)	(0.37)	(2.90***)	(2.57**)	
Panel B: 98 diversified portfolios								
Model	$\hat{\alpha}$	$\hat{\lambda}_{beta}$	$\hat{\lambda}_{size}$	$\hat{\lambda}_{value}$	$\hat{\lambda}_{mom}$	$\hat{\lambda}_{profit}$	$\hat{\lambda}_{investment}$	R^2
FF3	0.42	0.21	0.07	0.15				0.05
	(2.38**)	(0.93)	(0.60)	(1.27)				
C4	0.45	0.18	0.07	0.13	0.43			0.15
	(2.54**)	(0.83)	(0.57)	(1.12)	(2.19**)			
FF5	0.37	0.26	0.10	0.14		0.23	0.19	0.18
	(2.07**)	(1.21)	(0.82)	(1.22)		(2.50**)	(2.24**)	
FF6	0.53	0.11	0.09	0.15	0.42	0.22	0.17	0.30
	(2.92***)	(0.51)	(0.79)	(1.33)	(2.09**)	(2.44**)	(2.00**)	
Panel C: 286 anomaly portfolios								
Model	$\hat{\alpha}$	$\hat{\lambda}_{beta}$	$\hat{\lambda}_{size}$	$\hat{\lambda}_{value}$	$\hat{\lambda}_{mom}$	$\hat{\lambda}_{profit}$	$\hat{\lambda}_{investment}$	R^2
FF3	0.30	0.20	-0.11	0.15				0.04
	(17.92***)	(1.13)	(-0.57)	(1.06)				
C4	0.28	0.01	-0.13	0.15	0.24			0.07
	(17.56***)	(0.08)	(-0.74)	(1.07)	(1.18)			
FF5	0.28	0.20	-0.02	0.08		0.16	0.21	0.05
	(18.11***)	(1.19)	(-0.13)	(0.58)		(1.14)	(2.04**)	
FF6	0.26	0.01	-0.03	0.13	0.30	0.17	0.16	0.08
	(17.97***)	(0.07)	(-0.17)	(0.93)	(1.45)	(1.38)	(1.61)	

Note: Asterisks on t-statistics in parentheses indicate the level of statistical significance: *–10 percent, **–5 percent, and ***–1 percent.

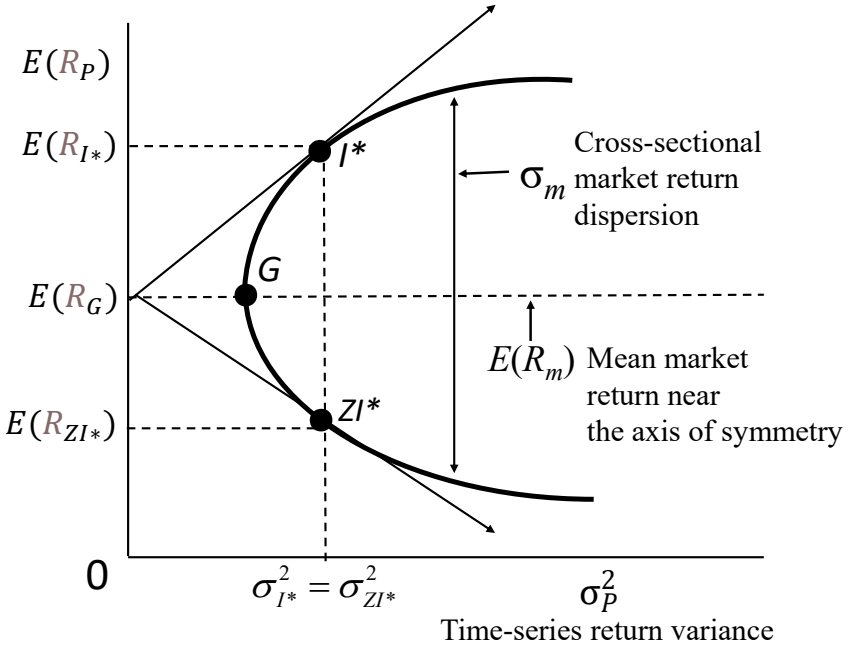


Figure 1: Illustration of the theoretical ZCAPM.

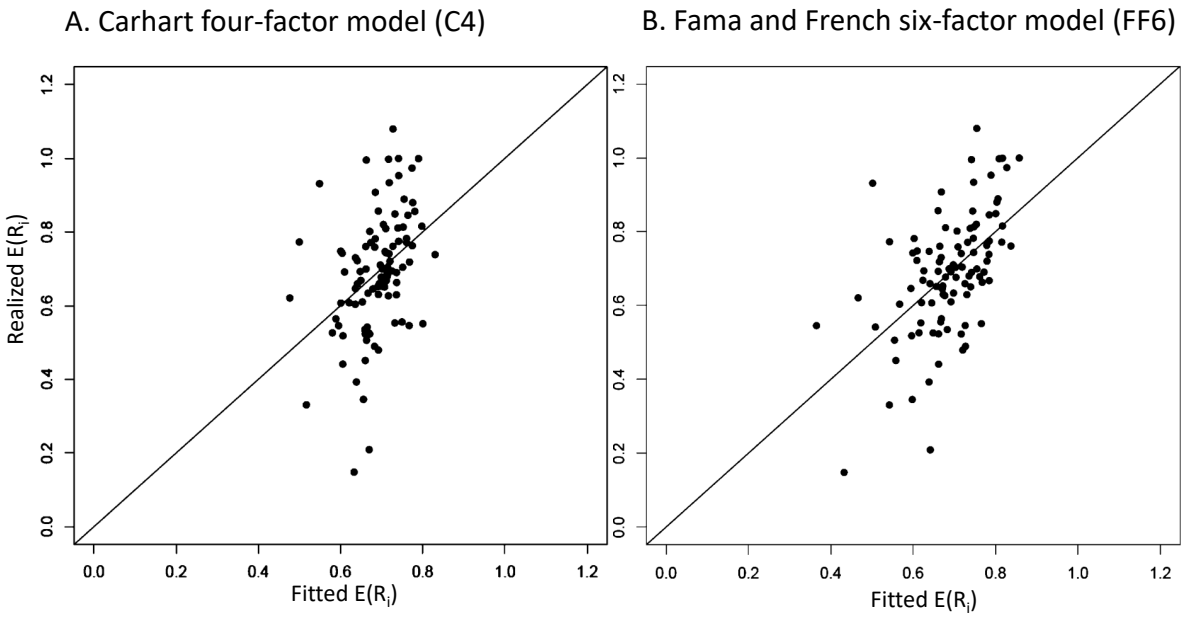


Figure 2: Out-of-sample average mispricing errors for Carhart and Fama and French multifactor models using 98 diversified portfolios.

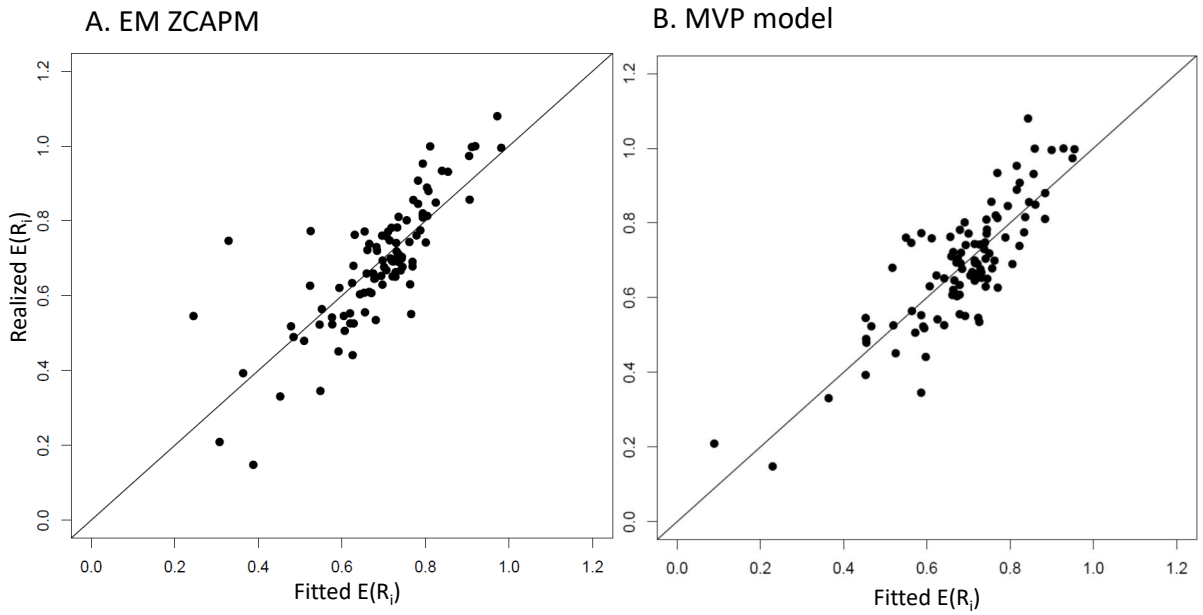


Figure 3: Out-of-sample average mispricing errors for EM ZCAPM and MVP model using 98 diversified portfolios.

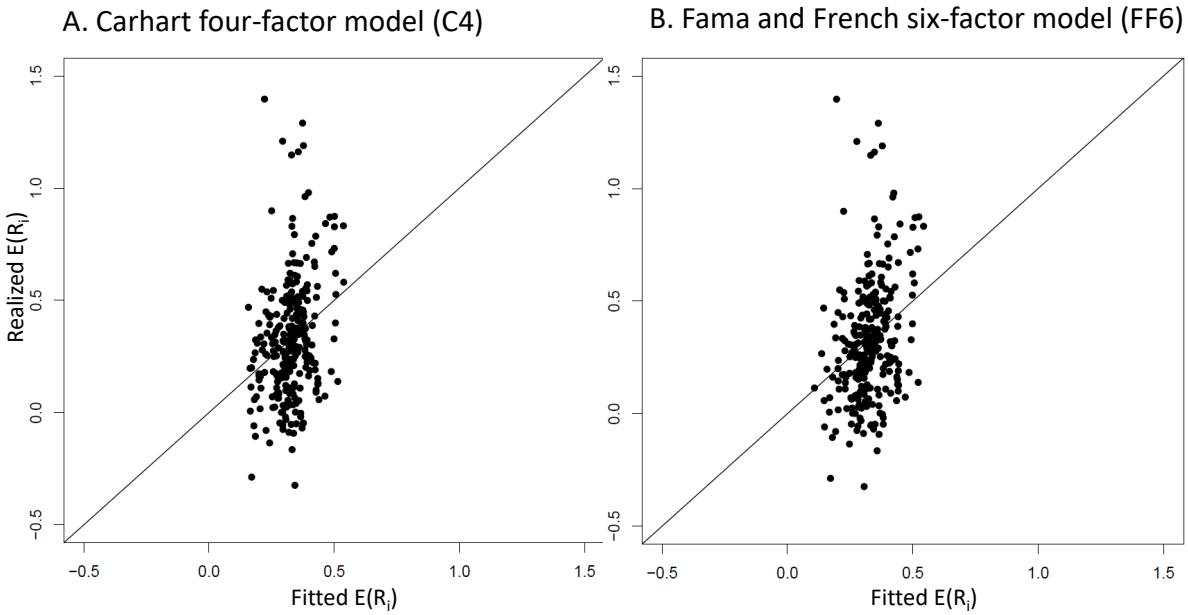


Figure 4: Out-of-sample average mispricing errors for Carhart and Fama and French multifactor models using 286 anomaly portfolios.

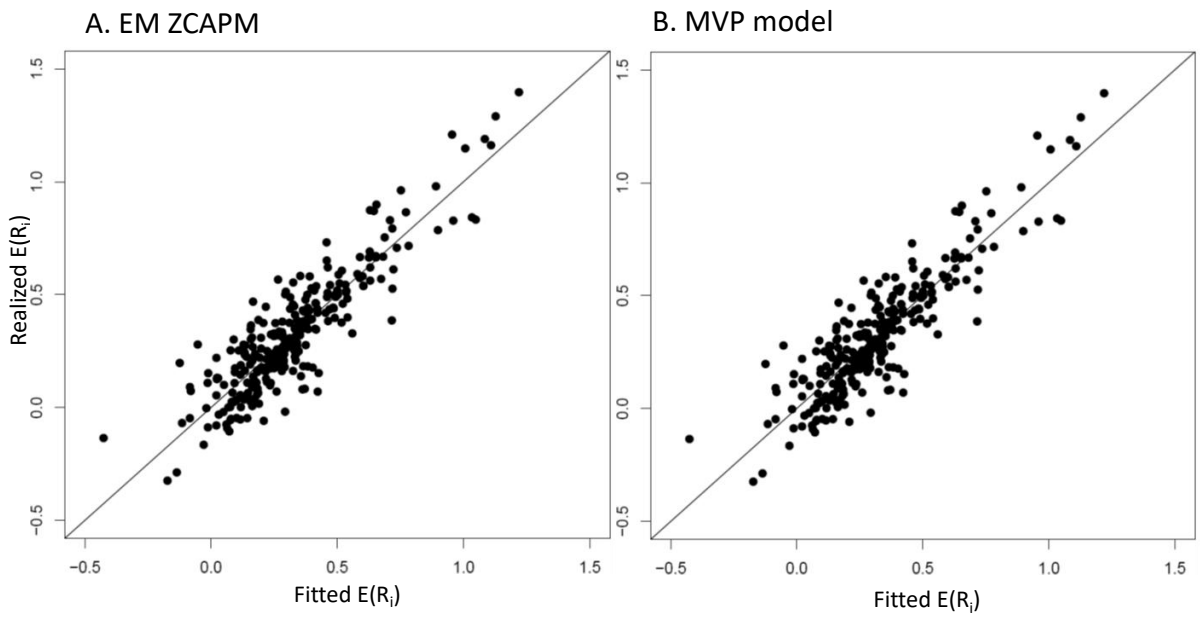


Figure 5: Out-of-sample average mispricing errors for EM ZCAPM and MVP model using 286 anomaly portfolios.